Even transitive markings on eleven points

Human Compression project

We apply the complete companion paper *Finite marked subgroups by successive stabilizers.* Our target is Fernandes–Piedade–Reade, *String C-group representations of transitive groups: a case study with degree 11*, arXiv:2302.11943v3, 1 November 2023, Theorem 8.8. Their proof uses permutation graphs and fracture graphs. The proof here uses a complete finite enumeration with a written coverage argument. It also recovers their principal permutation marking list in Proposition 8.4 and its ten orbit-size entries.

Theorem 1 (Fernandes–Piedade–Reade)**.** *Let $H \leq A_{11}$ be transitive and have a string C-group representation of rank four or five. Its rank is four, its group is isomorphic to* $\mathrm{PSL}_2(11)$*, and its abstract regular polytope is the self-dual* 11*-cell.*

Proof. Search all C-markings in the specified ambient A_{11} through rank five, modulo simultaneous conjugation by S_{11}. The companion theorem proves coverage and uniqueness, including markings generating proper transitive subgroups. The necessary completion test discards only prefixes whose remaining pool cannot generate a transitive group. Thus every marking in the statement must appear in the final transitive list.

The complete execution of `enumerate_even.g` visits

$$1,\ 2,\ 78,\ 2634,\ 147,\ 20$$

prefix classes at lengths zero through five, discarding 2497 branches by the necessary transitivity test. There are 130 transitive markings of rank three, exactly two of rank four, and none of rank five. Both rank-four groups have order 660, are explicitly isomorphic to $\mathrm{PSL}_2(11)$, and have type $\{3, 5, 3\}$. The full recorded tuples, their group computations and the complete search are required evidence, not optional tests against the predicted count.

The two rank-four tuples are equivalent after reversing one and conjugating points; as abstract ordered markings they are also isomorphic. The exact generator homomorphism sending either marking to its reverse is bijective. Their common abstract marking is the 11-cell: it agrees with source Proposition 8.4(5), whose literal permutations are printed below. The already acquired projective classification identifies this marking with the conventional 11-cell and supplies its geometric interpretation. Its full common paper and application, not only the theorem statement, remain retained inputs. The group–polytope correspondence gives the claimed uniqueness and abstract self-duality. □

Literal source markings. For a graphical statement there is an additional identification obligation. Number vertices in each row of the source diagram from left to right, proceeding from the top row to the bottom. The five source markings on physical page 23 become

$$\begin{aligned}
T_1 &= ((23)(45)(67)(89),\ (12)(34)(78)(9\,10), && (28)(39)(67)(10\,11)),\\
T_2 &= ((12)(34)(67)(89),\ (23)(45)(78)(9\,10), && (16)(27)(34)(10\,11)),\\
T_3 &= ((12)(34)(89)(10\,11),\ (23)(45)(78)(9\,10), && (1\,10)(2\,11)(34)(56)),\\
T_4 &= ((12)(67)(89)(10\,11),\ (23)(45)(78)(9\,10), && (16)(27)(34)(10\,11)),\\
T_5 &= ((12)(56)(89)(10\,11),\ (23)(45)(78)(10\,11), && (34)(56)(8\,10)(9\,11),\\
&\qquad (47)(58)(69)(10\,11)).
\end{aligned}$$

Rows 1,2,4 use five top and six bottom vertices; row 3 uses six top and five bottom vertices; row 5 uses rows of sizes six, three and two. Parentheses here denote disjoint cycles, not multiplication of adjacent decimal digits.

Proposition 2 (The source permutation list)**.** *Up to point conjugation and reversal, the transitive* $\mathrm{PSL}_2(11)$ *markings of ranks three and four are exactly* $T_1, \ldots, T_5$*. Up to abstract marked-group isomorphism and reversal they give four polytopes, of types* $\{5,5\}$*,* $\{5,6\}$*,* $\{6,6\}$ *and* $\{3,5,3\}$*.*

Proof. The complete transitive rank-three census consists of 122 markings generating A_{11} and eight generating groups of order 660, explicitly identified with $\mathrm{PSL}_2(11)$. The latter eight form four reversal pairs under point conjugation. The rank-four list forms one such pair. Direct comparison with the five displayed source tuples identifies all five classes, with no additional class. A separate full-subset intersection test checks each source tuple and every recorded transitive tuple.

For abstract equivalence, generator homomorphisms show that T_2 and T_3 represent the same ordered marking, whereas their point representations differ. The other three markings are abstractly self-dual; T_2, T_3 have type $\{5,6\}$ and are interchanged with the ordered dual type $\{6,5\}$ on reversal. Distinct unordered types separate the four abstract polytopes. All these homomorphism and conjugacy comparisons are retained in the complete finite evidence. □

Deleting the first mark, and then doing the same after reversal, gives the following orbit sizes, exactly source physical page 24:

Source row	Suffix orbits	Reversed suffix orbits
1	$1,5,5$	$1,5,5$
2	$2,3,6$	$1,5,5$
3	$2,3,6$	$1,5,5$
4	$2,3,6$	$2,3,6$
5	$1,10$	$5,6$

The distinction between abstract and permutation self-duality is essential: none of these five displayed permutation markings is conjugate in S_{11} to its reverse, although three underlying abstract markings are self-dual.

Scope and attribution. Theorem 1 is the source's main result, not a new conjecture answer. The rank-three and permutation classifications retain the source's attribution to the earlier regular-map atlases; the 11-cell retains its attribution to Grünbaum and Coxeter. The complete finite search replaces the source's case analysis for the stated principal contract. It does not assert recovery of every independent auxiliary statement about arbitrary permutation graphs, nor of the source appendix's 52 claimed independent generating tuples. These are separately disposed in the source map, and are not premises of this classification. No claim of literal theorem-set equivalence, numerical compression or new Lean formalization is made.